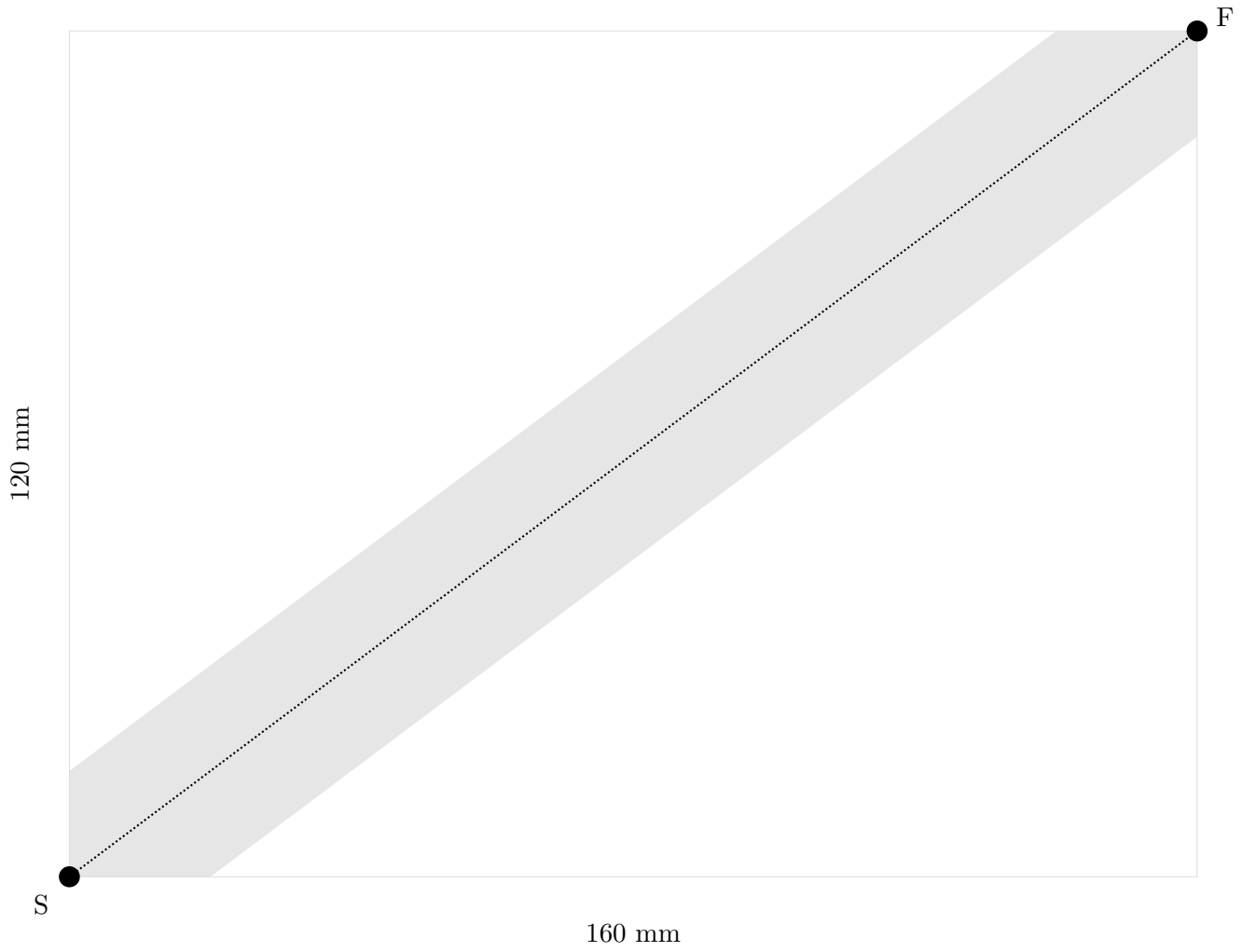
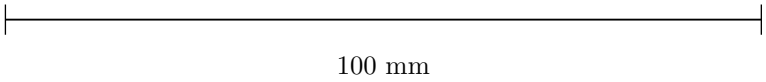
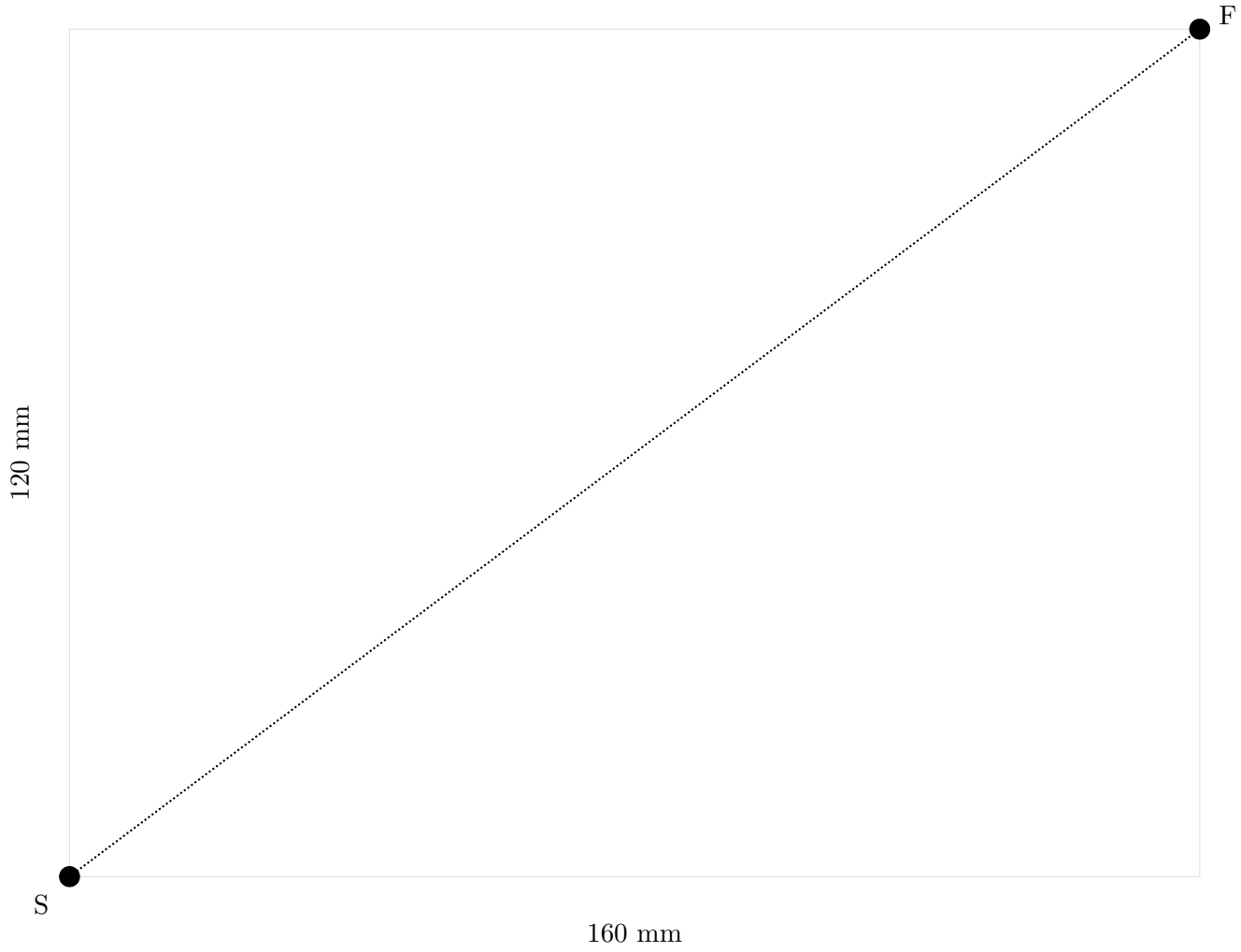
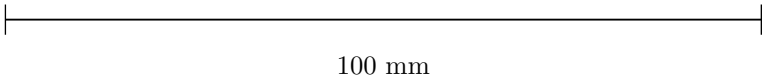
Problem 4: Find the fewest turns possible for an S-to-F staircase inside the gray strip. Its pieces may follow the strip's boundary. Explain why fewer turns cannot work.



The gray strip reaches 15 mm vertically above and below the dotted diagonal, clipped by the rectangle.



Problem 5: Allow leftward pieces as well as rightward and upward pieces. Make two very different S-to-F paths of length 360 mm. Could such a path be shorter than 280 mm? Explain your answer.



A straight diagonal is 200 mm long.